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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	1006901	4	Acidity, Dielectric, Color, Visualization and Water Content
2.	TRAFO 1 ESP UNIT 1	10937	5	Acidity, Dielectric, Color, Visualization and Water Content
3.	TRAFO 2 ESP UNIT 1	10035	6	Acidity, Dielectric, Color, Visualization and Water Content
4.	TRAFO 3 ESP UNIT 1	10031	7	Acidity, Dielectric, Color, Visualization and Water Content
5.	TRAFO 4 ESP UNIT 1	10034	8	Acidity, Dielectric, Color, Visualization and Water Content
6.	MAIN TRAFO UNIT 3	1122170	9	Acidity, Dielectric, Color, Visualization and Water Content
7.	UAT 3	1035082	10	Acidity, Dielectric, Color, Visualization and Water Content
8.	TRAFO 1 ESP UNIT 3	0073084208	11	Acidity, Dielectric, Color, Visualization and Water Content
9.	TRAFO 2 ESP UNIT 3	0070084208	12	Acidity, Dielectric, Color, Visualization and Water Content
10.	TRAFO 3 ESP UNIT 3	0072084208	13	Acidity, Dielectric, Color, Visualization and Water Content
11.	TRAFO 4 UNIT 3	0071084208	14	Acidity, Dielectric, Color, Visualization and Water Content
12.	ASH HANDLING	10820108	15	Acidity, Dielectric, Color, Visualization and Water Content
13.	ASH HANDLING	10820107	16	Acidity, Dielectric, Color, Visualization and Water Content

Note:

* OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	1006901	4

A) Analysis

- **Acidity** of 0.017mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 70kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 8ppm is **ACCEPTABLE** while moisture in solid insulation of 1.04% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.059mgKOH/g is **QUESTIONABLE**. Together with previous low IFT value, these two parameters indicate that the oil is contaminated with oxidation byproducts such as acid and dissolved sludge. If left unattended, these oxidation byproducts will eventually deposit on the core of the transformer, causing insufficient heat dissipation, as well as degradation of the paper insulation.
- **Dielectric** of 57kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 20ppm is **QUESTIONABLE** while moisture in solid insulation of 2.60% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Most of the parameters show that the liquid insulation is in bad condition. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.045mgKOH/g is **ACCEPTABLE**. While Acidity is still within acceptable levels, it is approaching the questionable range. Together with earlier oil analysis for the IFT, these two parameters indicate the presence of oxidation byproducts in the oil. Oil oxidation byproducts promotes more acids to be produced by the present acids. Therefore, removal of these contaminations is highly recommended.
- **Dielectric** of 25kV is **QUESTIONABLE**. Low dielectric value for insulating oil is not conducive for normal operation of a transformer unit
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 27ppm is **UNACCEPTABLE** while moisture in solid insulation of 3.91% M/dw is graded as **E**. For this transformer, it is recommended to maintain the water content less than 20ppm. High moisture content in the transformer will accelerate oxidation process and cause dielectric breakdown.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation. Most of the parameters show that the liquid insulation is in bad condition. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.042mgKOH/gm is **ACCEPTABLE**. Close monitoring is advised as low IFT detected on previous oil analysis indicate that there are contaminant in the transformer oil that is weakening the molecular bonds of the insulating oil.
- **Dielectric** of 77kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 16ppm is **ACCEPTABLE** while moisture in solid insulation of 2.08% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Retest **Acidity, Interfacial Tension, and Liquid Power Factor** in 6 months to monitor their levels.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**. Base on earlier oil analysis for the IFT value, low IFT level indicates the oil is containing contaminants that had weakened the molecular bonds of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 88kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 15ppm is **ACCEPTABLE** while moisture in solid insulation of 1.95% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
6.	MAIN TRAF0 UNIT 3	1122170	9

A) Analysis

- **Acidity** of 0.014mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 77kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 17ppm is **ACCEPTABLE** while moisture in solid insulation of 1.21% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
7.	UAT 3	1035082	10

A) Analysis

- **Acidity** of 0.025mgKOH/gm is **ACCEPTABLE**.
- **Dielectric** of 88kV is **ACCEPTABLE**.
- **Colour** of 0.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 14ppm is **ACCEPTABLE** while moisture in solid insulation of 1.33% M/dw is graded as **B**. For this transformer, it is recommended to maintain the water content less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	0073084208	11

A) Analysis

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE**. Base on earlier oil analysis for the IFT value, low IFT level indicates the oil is containing contaminants that had weakened the molecular bonds of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 47kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 21ppm is **QUESTIONABLE** while moisture in solid insulation of 2.00% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 20ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.112mgKOH/g is **UNACCEPTABLE**. The result shows that the condition of the transformer oil is deteriorating. Visible sludge is already formed in and around the transformer core. These oxidation products are in fact attacking the paper insulation and tearing them apart. Loss of the paper's mechanical and dielectric strength signals the end of the useful life of the transformer.
- **Dielectric** of 28kV is **QUESTIONABLE**. This is reflected from the high moisture content detected in the transformer oil.
- **Colour** of 3.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 38ppm is **UNACCEPTABLE** while moisture in solid insulation of 3.61% M/dw is graded as **E**. For this transformer, it is recommended to maintain the water content less than 20ppm. High moisture content will cause dielectric failure and act as a catalyst for oxidation process.

B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit to improve Acidity, IFT, Dielectric, and Water Content. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as excess moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE**. Base on earlier oil analysis for the IFT value, low IFT level indicates the oil is containing contaminants that had weakened the molecular bonds of the oil. Removal of these contaminations is highly recommended.
- **Dielectric** of 66kV is **ACCEPTABLE**.
- **Colour** of 1.5 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 18ppm is **ACCEPTABLE** while moisture in solid insulation of 1.89% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 30ppm.

B) Recommendation

- It is still recommended to perform **Transformer Oil Regeneration** on this unit as per previous recommendation. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.022mgKOH/gm is **ACCEPTABLE**. Close monitoring is advised as low IFT detected on previous oil analysis indicate that there are contaminant in the transformer oil that is weakening the molecular bonds of the insulating oil.
- **Dielectric** of 60kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 15ppm is **ACCEPTABLE** while moisture in solid insulation of 1.42% M/dw is graded as **C**. For this transformer, it is recommended to maintain the water content less than 20ppm.

B) Recommendation

- Retest **Acidity, Interfacial Tension, and Liquid Power Factor** in 6 months to assess if the insulating oil deteriorate to a level that require corrective action to improve the oil quality.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
12.	ASH HANDLING	10820108	15

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 56kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 21ppm is **ACCEPTABLE** while moisture in solid insulation of 2.73% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content less than 30ppm.

B) Recommendation

- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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No.	Unit No.	Serial Number	TC
13.	ASH HANDLING	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE**.
- **Dielectric breakdown voltage** of 40kV is **ACCEPTABLE**.
- **Colour** of 1.0 utilizing ASTM D1500 standard is **ACCEPTABLE**.
- **Visualization** of transformer oil is CLEAR, thus **ACCEPTABLE**.
- **Water Content** in oil of 32ppm is assessed as **QUESTIONABLE**.

Meanwhile, Water Content in solid insulation of 3.04% M/dw is graded as **D**. For this transformer, it is recommended to maintain the water content below 30ppm. High level of moisture content will act as catalyst to the oxidation process.

B) Recommendation

- Retest **Moisture** and **Dielectric** in 6 months to ascertain if corrective oil work is required to remove the moisture to enable optimum operating condition for the transformer.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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3.0 Summary of Our Comments & Recommendations:

No.	TX Unit No.	Serial No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	1006901	4	• Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	• Perform OIL REGENERATION
3.	TRAFO 2 ESP UNIT 1	10035	6	• Perform OIL REGENERATION
4.	TRAFO 3 ESP UNIT 1	10031	7	• Retest 6 months: Acidity, IFT and Liquid Power Factor • Other parameters: Periodic condition monitoring
5.	TRAFO 4 ESP UNIT 1	10034	8	• Perform OIL REGENERATION
6.	MAIN TRAFO UNIT 3	1122170	9	• Periodic condition monitoring
7.	UAT 3	1035082	10	• Periodic condition monitoring
8.	TRAFO 1 ESP UNIT 3	0073084208	11	• Perform OIL REGENERATION
9.	TRAFO 2 ESP UNIT 3	0070084208	12	• Perform OIL REGENERATION
10.	TRAFO 3 ESP UNIT 3	0072084208	13	• Perform OIL REGENERATION
11.	TRAFO 4 UNIT 3	0071084208	14	• Retest 6 months: Acidity, IFT, and Liquid Power Factor • Other parameters: Periodic condition monitoring
12.	ASH HANDLING	10820108	15	• Periodic condition monitoring

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No.	TX Unit No.	Serial No.	TC	Recommendation
13.	ASH HANDLING	10820107	16	• Retest 6 months: Dielectric, and Water Content • Other parameters: Periodic condition monitoring

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